Power and resource tradeoffs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Compare model-specific detection estimates with resource costs

Distinguish supplied alternative-trial frequencies, per-study cost and unmeasured null error.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Compare model-specific detection estimates with resource costs

> Distinguish supplied alternative-trial frequencies, per-study cost and unmeasured null error.

Learning goals

I can explain the **mechanism** and terminology of **power** and resource tradeoffs.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Pilot experiments. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **power** and resource tradeoffs using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **power** and resource tradeoffs

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Distinguish supplied alternative-trial frequencies, per-study cost and unmeasured null error.

Start with the research problem

Power is the probability that a specified test rejects under a specified alternative effect and design. A finite set of simulations can estimate that probability, with simulation uncertainty. This lesson supplies counts; it does not run those simulations. Larger sample size may improve detection under the stated assumptions while costing more, but no universal sample-size rule follows.

Keep the effect, variability, unit independence, test and resource model visible.

Original concept explanation

Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Complete case inputs, rules and limits

Planning item	Complete main example
Alternative model	Fixed supplied alternative and test; model details not provided for reproduction
Trial counts	One hundred alternative simulations per design, stipulated
Small design	n = 20 independent units; 62 detections
Large design	n = 40 independent units; 83 detections
Unit cost	Three credits per independent unit
Per-study cost	n × 3, not the total cost of all one hundred simulation trials
Unknown quantities	Null false-positive rate, simulation uncertainty interval, real-study power and actual compute cost
Separate picture	Different supplied counts: 60/100 and 80/100, unit cost two resource units

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Complete main-case evidence ledger

Main quantity	n = 20	n = 40
Supplied detections / alternative trials	62/100	83/100
Estimated detection frequency	62%	83%
Independent units per study	20	40
Per-unit cost	3 credits	3 credits
Per-study cost	20 × 3 = 60 credits	40 × 3 = 120 credits
Large minus small	21 percentage points more observed detection	60 extra credits per study
Null error and actual-study guarantee	Not supplied	Not supplied

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Power and resource tradeoffs — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits.	Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.	Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Change the assumed effect or variability but keep the sample sizes.

▼ **Check after predicting**

Prediction: The detection fractions need new simulation; the resource costs stay 40 and 80.

Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits.

Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

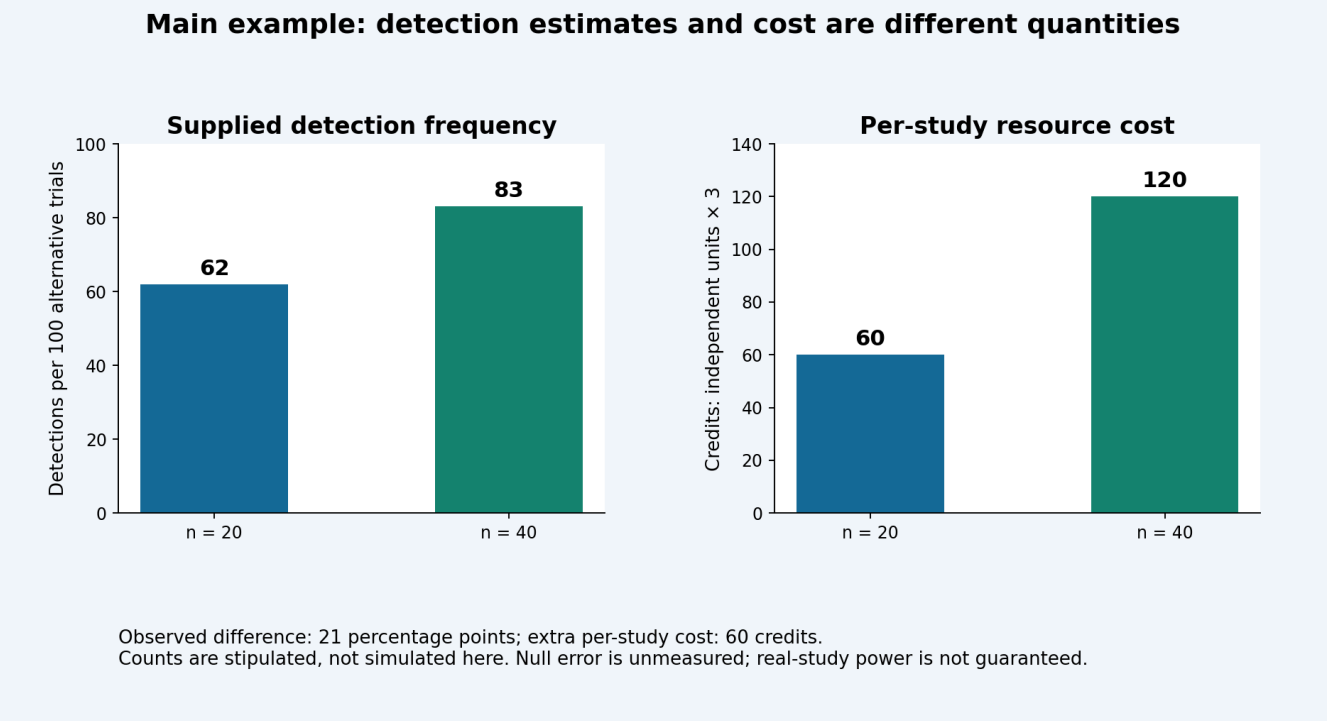
A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

The separate picture uses one hundred stipulated alternative trials per design, with sixty detections at n = 20 and eighty at n = 40. Its unit cost is two resource units, unlike the main example’s three credits. Estimated detection frequencies are 60% and 80%; per-study costs are forty and eighty resource units. The extra forty resource units accompany a twenty-percentage-point observed difference.

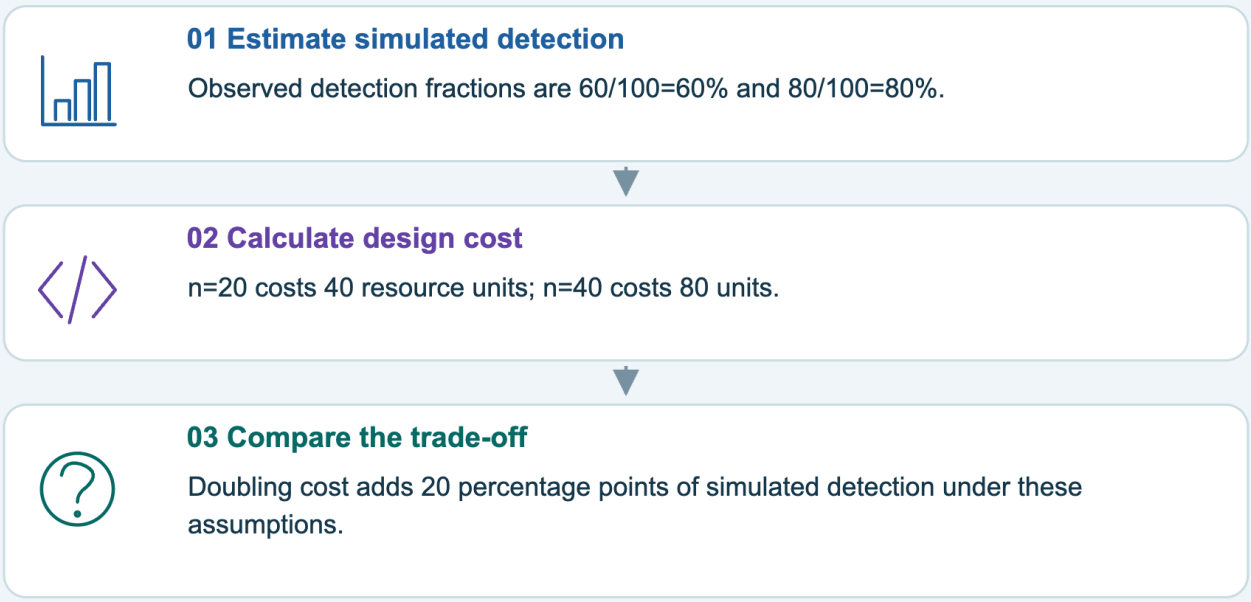
Neither is a measured effect on real pupils. Changing the assumed effect or variability may change power even at the same sample sizes, but no replacement detection counts are supplied. Null false-positive error needs separate null trials or appropriate theory.

Read the labelled representation



AI ACADEMY / AI-7/23 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Estimate simulated detection	Observed detection fractions are 60/100=60% and 80/100=80%.
2. Calculate design cost	n=20 costs 40 resource units; n=40 costs 80 units.
3. Compare the trade-off	Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Complete picture evidence and limits

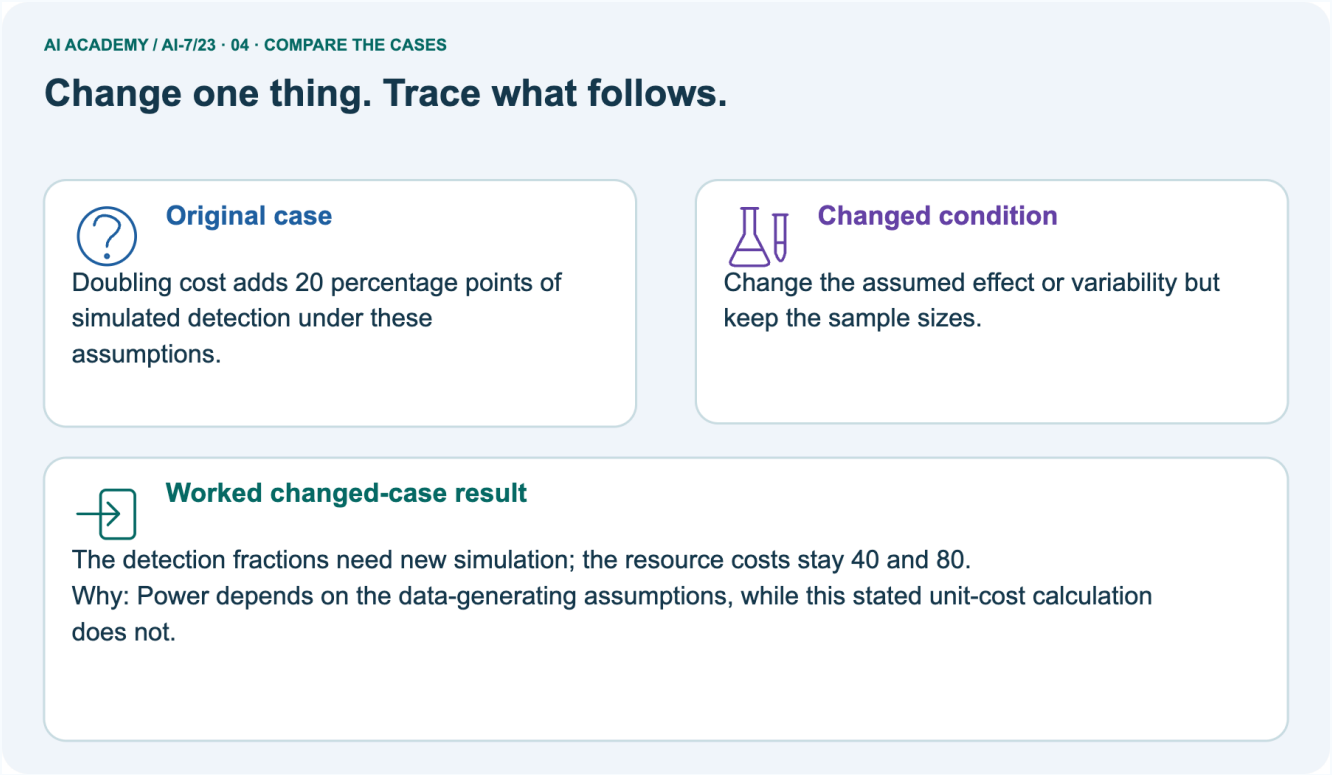
Separate picture quantity	n = 20	n = 40
Detections / supplied trials	60/100	80/100
Detection frequency	60%	80%
Cost per independent unit	2 resource units	2 resource units

Separate picture quantity	n = 20	n = 40
Per-study cost	40 resource units	80 resource units
Large minus small	20 percentage points more detection	40 extra resource units
Changed effect/variability	New frequency unknown	New frequency unknown
What was actually run here	No simulations	No real study

Predict the changed condition

- Change one thing: Change the assumed effect or variability but keep the sample sizes. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not executed simulation or **evidence** about pupils.

- Read the labelled picture: Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

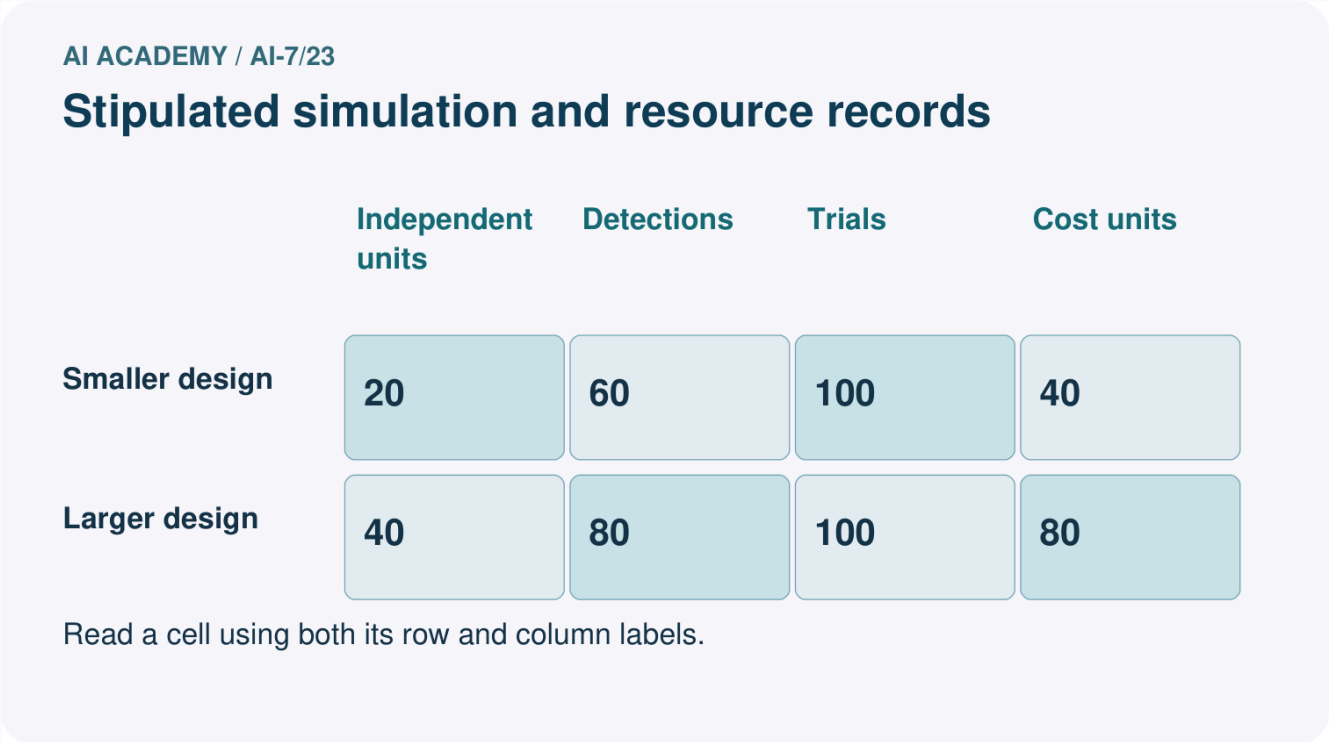
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Power and resource tradeoffs

The facts we will use



AI ACADEMY / AI-7/23

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/23

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are 60/100=60% and 80/100=80%.



2. Calculate design cost

n=20 costs 40 resource units; n=40 costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Power/budget probe	Checked interpretation
62 detections out of 100 alternative trials	62% estimate, not known exact power
No null trials	False-positive frequency is unmeasured
Per-study cost versus all simulation work	Different denominators; do not confuse them
More units under fixed assumptions	Compare supplied frequencies and cost; no universal rule
Effect or variability changes	Recompute under the new model; old power need not transfer
Different unit IDs	Does not itself establish independent sampling
Higher observed detection rate	Not a guarantee the next study succeeds
Simulation model unspecified	Do not pretend supplied counts can reproduce the simulation

Write and justify the repair

- A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example "**Power** and resource tradeoffs". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Power** and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

> **My explanation using a pictured step and the given facts**

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Predict main detection fractions 62/100 and 83/100 and costs 20 × 3 and 40 × 3. The complete code summarizes supplied counts with exact fractions; it does not generate trials. Compare with the separate picture's 60/100 and 80/100 at unit cost two. Distinguish the per-study budget from hypothetical total work across one hundred trials; no actual billing or compute use is measured.

   Record missing model details, null trials and uncertainty rather than inventing them. Keep the independent n = 30 versus n = 60 task for a first attempt.

```
"""Summarize stipulated alternative-model simulation counts; do not simulate trials."""
from fractions import Fraction

def summarize(detections,trials,n,cost_per_unit):
 if any(type(x) is not int for x in (detections,trials,n,cost_per_unit)) or trials<1 or
n<1 or cost_per_unit<1 or not 0<=detections<=trials:
  raise ValueError('valid integer trial counts, sample size and positive unit cost
required')
 return
{'detection_fraction':Fraction(detections,trials),'per_study_cost':n*cost_per_unit,
         'stated_simulation_work_cost':trials*n*cost_per_unit}
main_small=summarize(62,100,20,3);main_large=summarize(83,100,40,3)
picture_small=summarize(60,100,20,2);picture_large=summarize(80,100,40,2)
assert main_small['per_study_cost']==60 and main_large['per_study_cost']==120
assert 100*(main_large['detection_fraction']-main_small['detection_fraction'])==21
assert picture_small['per_study_cost']==40 and picture_large['per_study_cost']==80
assert picture_large['detection_fraction']-
picture_small['detection_fraction']==Fraction(1,5)
```

```
print('Main summaries:',main_small,main_large)
print('Separate picture summaries:',picture_small,picture_large)
print('Per-study cost differs from total hypothetical work across 100 simulation trials;
the latter is arithmetic, not measured compute or billing.')
print('These are supplied finite-trial detection frequencies under a fixed alternative,
not executed simulations, exact known power or guaranteed real-study success.')
print('No null trials are supplied: false-positive rate is unknown. Changed
effect/variability requires new model-specific planning; no new detection frequency is
invented.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

Try the selected new case independently

> Compare both supplied detection fractions and per-study costs, then state the model and null-error limitations.

Complete independent task

One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled **power** and resource tradeoffs **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **power** and resource tradeoffs.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **power** and resource tradeoffs

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Power	The test's detection probability under a specified alternative
Simulation trial	A generated dataset under a declared model
Planning effect	The effect size used for design calculations

Use two terms to explain a step in this week’s example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Power and resource tradeoffs — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits.	Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.	Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

By the end, I can	How I will show it
Explain the mechanism and terminology of power and resource tradeoffs	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.
- Which labelled step matches this detail? Doubling cost adds 20 percentage points of simulated detection under these assumptions.
- 1. Compare the trade-off
- 2. Estimate simulated detection
- 3. Calculate design cost
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare the trade-off Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Changed-case question and optional hint

- Changed-case practice: Change the assumed effect or variability but keep the sample sizes. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The detection fractions need new simulation; the resource costs stay 40 and 80.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The detection fractions need new simulation; the resource costs stay 40 and 80.

Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Repair and retain the record

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

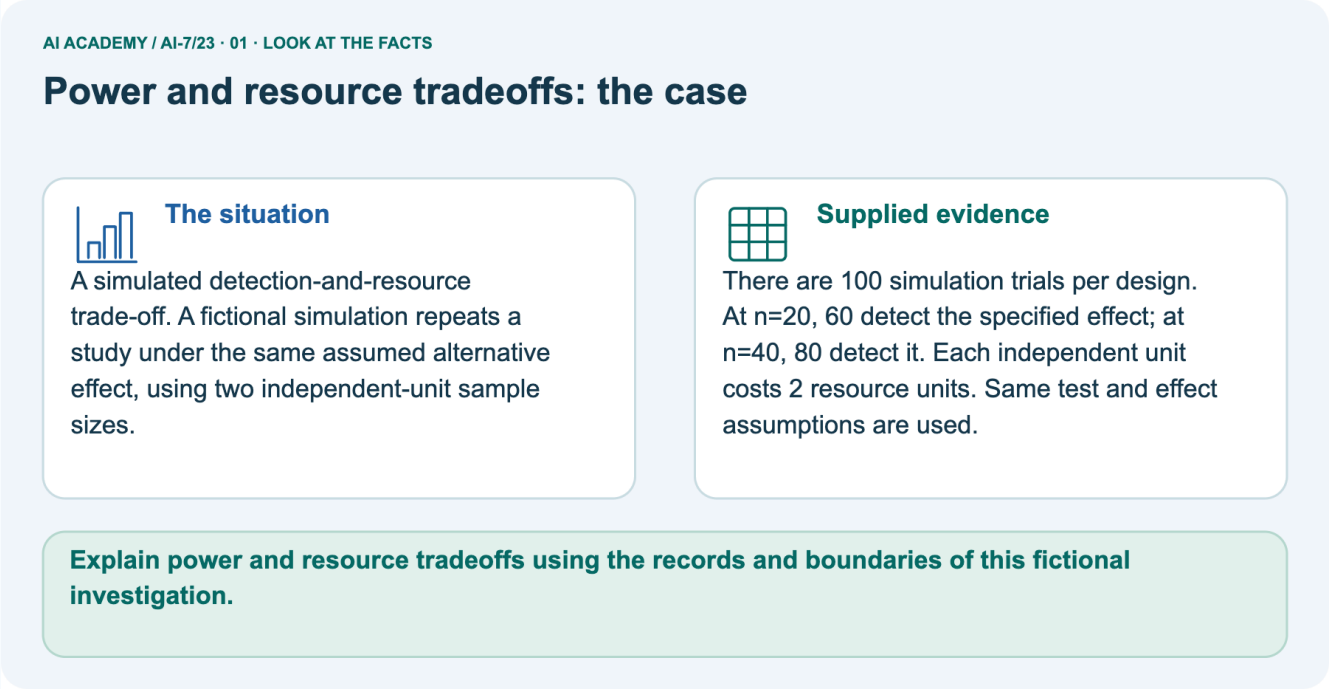
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Power and resource tradeoffs: the case



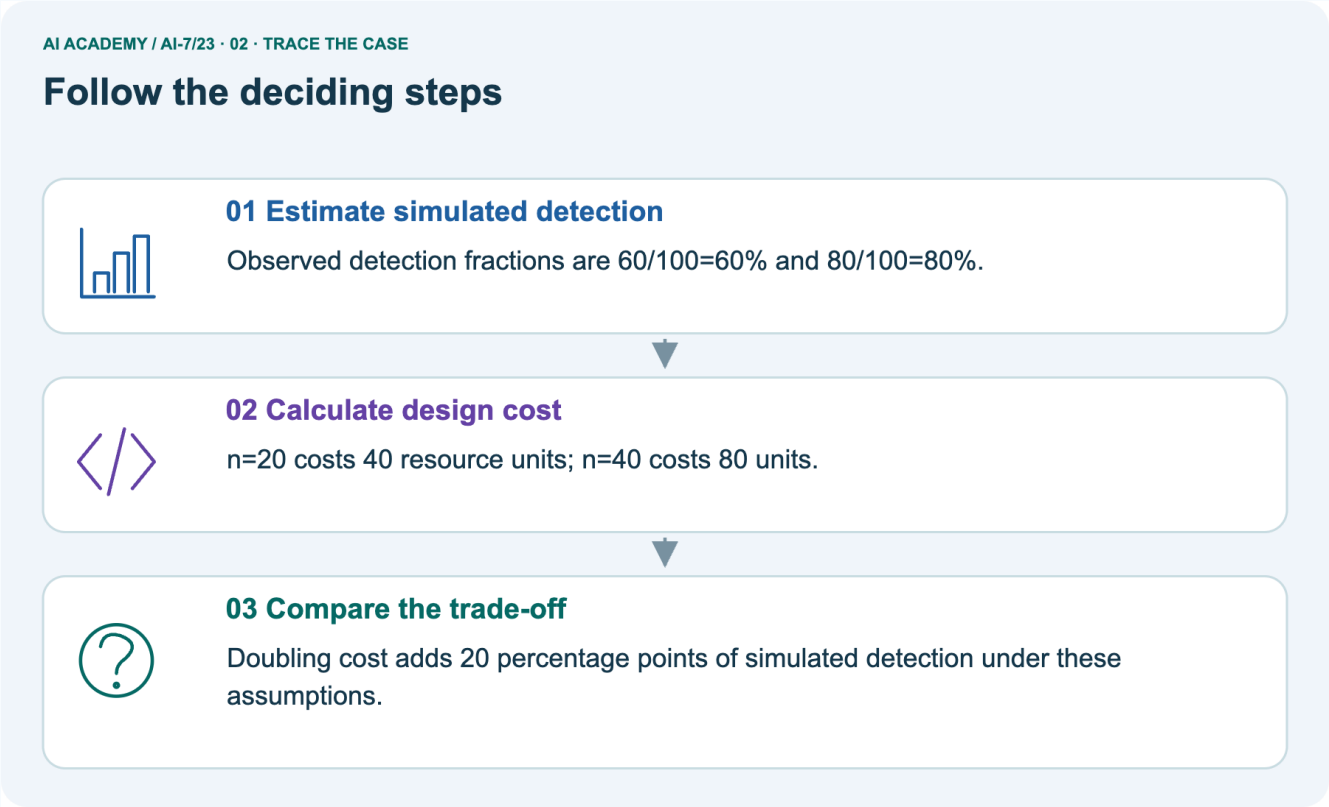
Follow the labelled picture with the matching explanation. Open the vector diagram

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied facts: There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

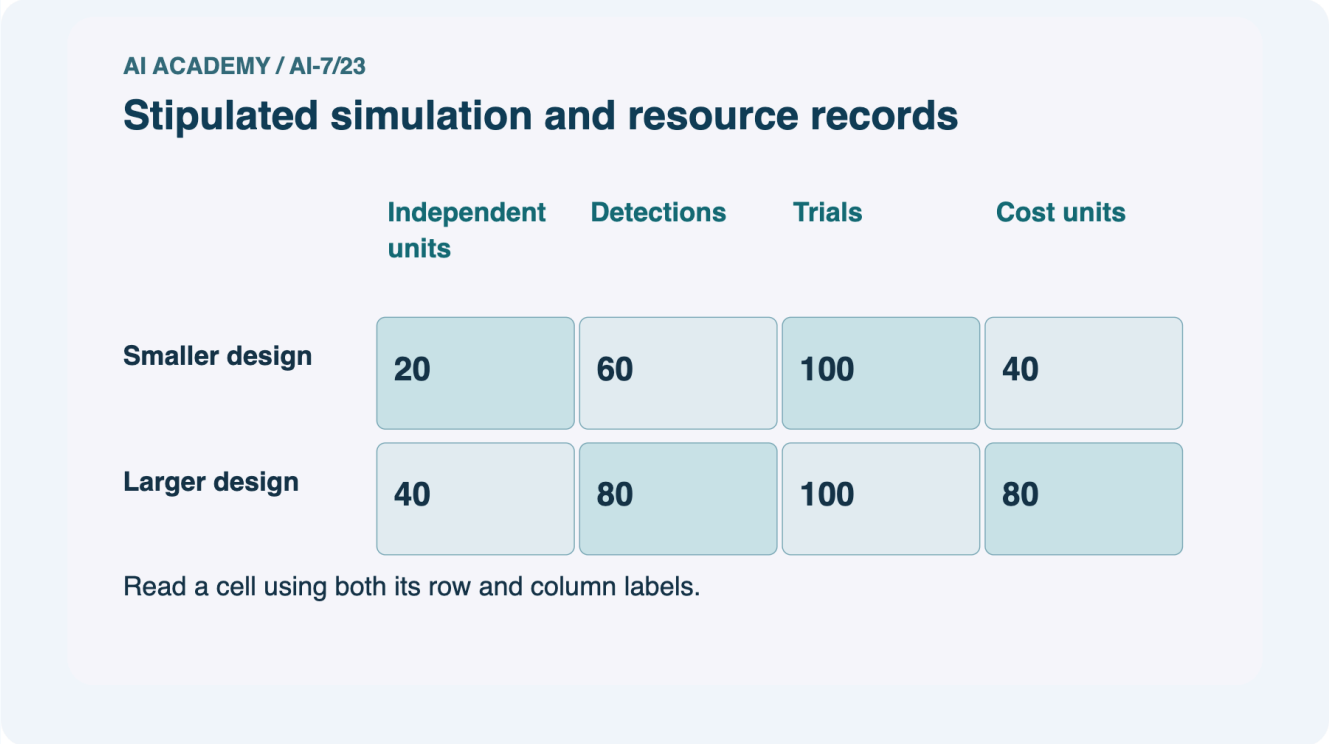


Follow the labelled picture with the matching explanation. Open the vector diagram

Estimate simulated detection: Observed detection fractions are 60/100=**60%** and 80/100=**80%**. → Calculate design cost: n=20 costs 40 resource units; n=40 costs 80 units. → Compare the trade-off: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Point to each numbered stage and explain which recorded **input** it uses.

03. Stipulated simulation and resource records

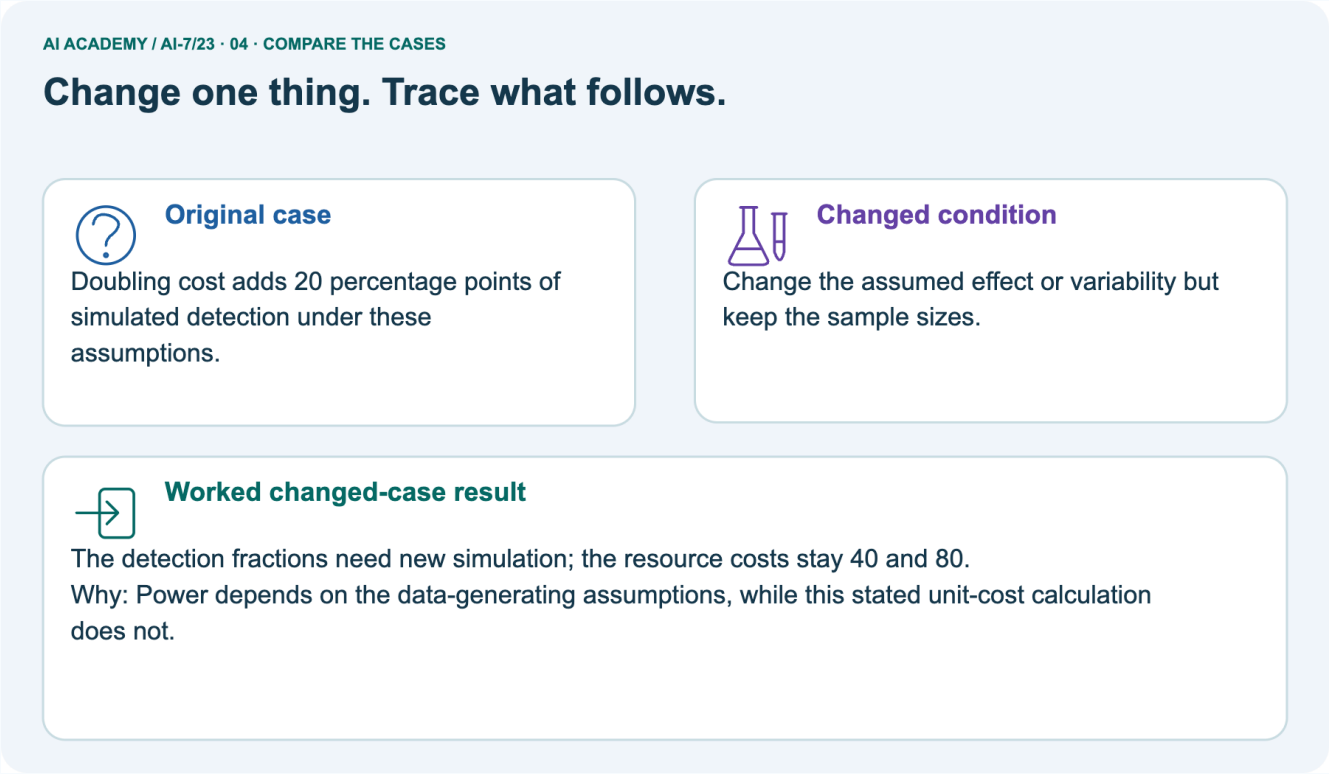


Follow the labelled picture with the matching explanation. Open the vector diagram

Stipulated simulation and resource records. Smaller design: Independent units = 20, Detections = 60, Trials = 100, Cost units = 40. Larger design: Independent units = 40, Detections = 80, Trials = 100, Cost units = 80.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



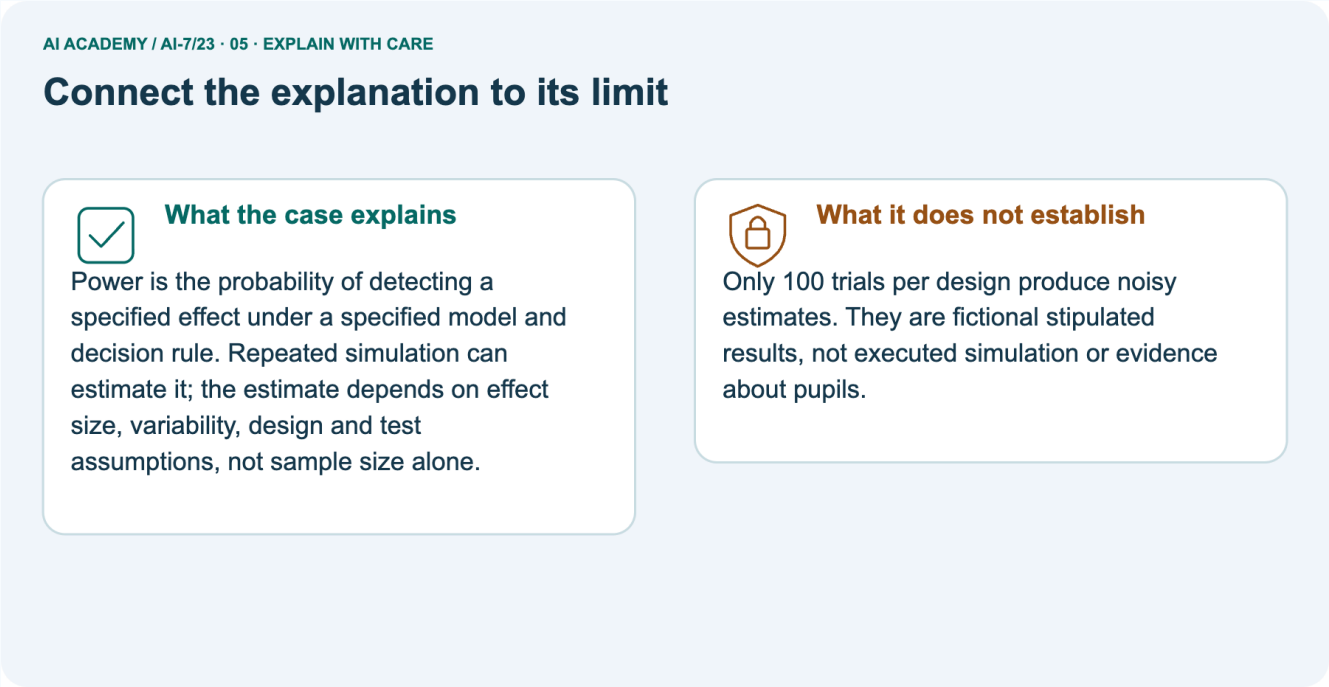
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Doubling cost adds 20 percentage points of simulated detection under these assumptions. Changed condition: Change the assumed effect or variability but keep the sample sizes.

Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

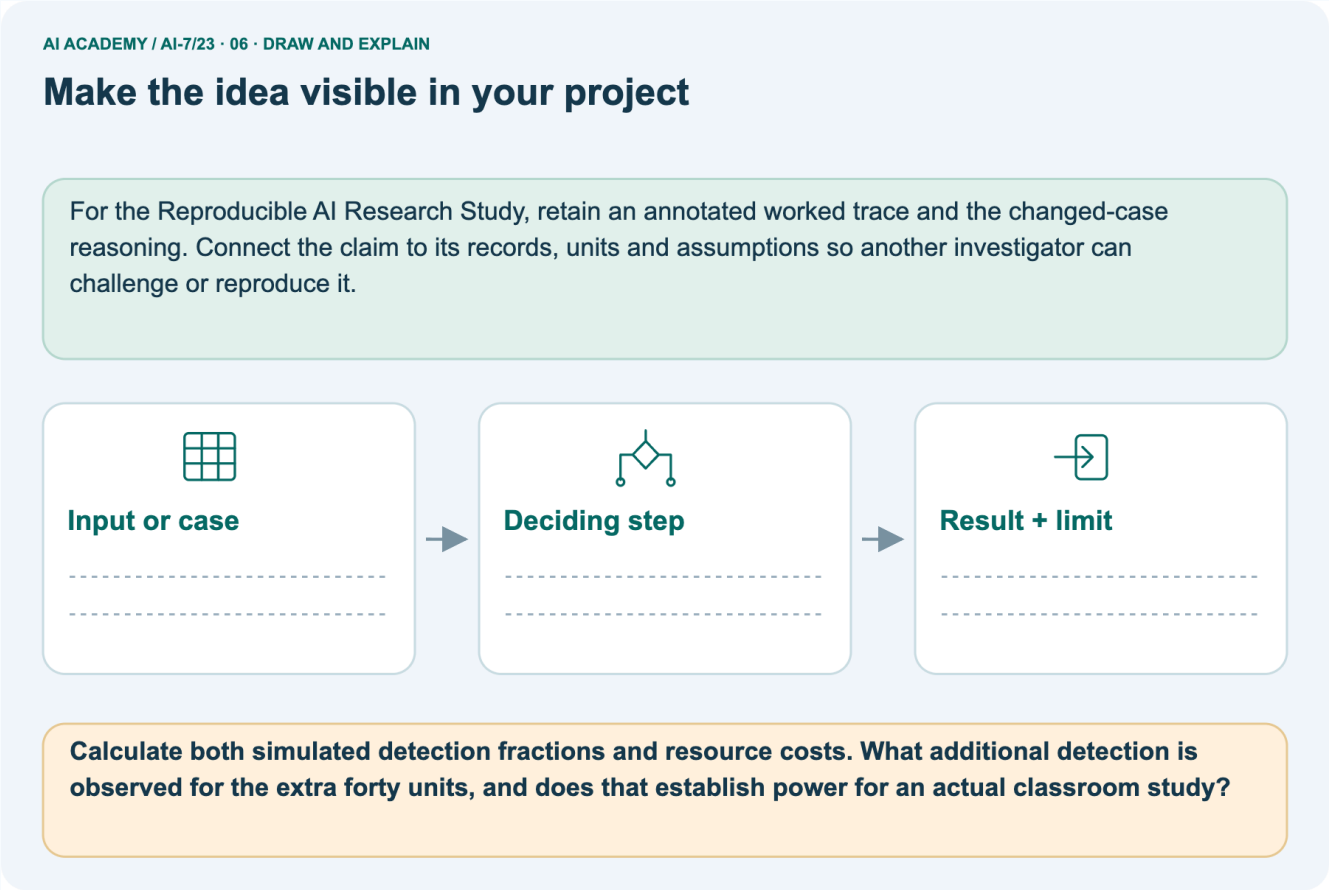
Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone. Limit: Only 100 trials per design produce noisy estimates.

They are fictional stipulated results, not executed simulation or **evidence** about pupils.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Power and resource tradeoffs: the case

02. Follow the deciding steps

03. Stipulated simulation and resource records

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.